

Lecture 9 (10/5)

Bias-variance trade off, confidence intervals and least squares

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Example

Last class we looked at the sample mean $\bar{Y}$ as an estimator both for Bernoulli and Normal. Is this unbiased and consistent?

Example

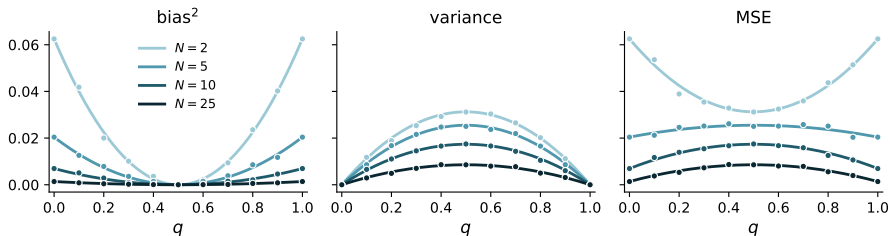
Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$. The natural estimator of q is $\hat{q} = \bar{Y}$. Thus, if every $Y_i = 0$, then $\hat{q} = 0$. This is problematic: If we flip the coin twice, this would lead us to conclude the coin can never land on heads. To correct for this, we pretend we have already seen two flips, one heads and one tails.

Laplace's rule of succession is the estimator

$$\hat{q}_L = \frac{S + 1}{N + 2}, \quad S = \sum_i Y_i.$$

Find the bias, variance and MSE of $\hat{q}_L$.

$$\text{bias} = \frac{1 - 2q}{N + 2}, \quad \text{se}(\hat{q}_L)^2 = \frac{Nq(1 - q)}{(N + 2)^2}$$



Curves are formulas above. Points are estimates from 1000 simulated replicates (see code on next slide).

```

N = 10
R = 1000 # number of replicates
qs = np.linspace(0, 1, 11)
for i in range(len(qs)):
    q = qs[i]
    q_hat = np.zeros(R)
    for r in range(R):
        S = np.random.binomial(N, q)
        # This also works:
        # ys = np.random.choice([0, 1], N, p=[1-q, q])
        # S = np.sum(ys)
        q_hat[r] = (S + 1) / (N + 2)
    bias2[i] = (np.mean(q_hat) - q)**2
    var[i] = np.var(q_hat)
    mse[i] = np.mean((q_hat - q)**2)

```

Each pass through the inner loop is one replicate of the data, so `q_hat` holds R draws from the sample distribution of $\hat{q}_L$. We repeat this for $N = 2, 5, 10, 25$.

Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}_\mu^2 = \frac{1}{N} \sum_i (Y_i - \mu)^2$$

Is this an unbiased estimator?

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Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is NOT known to us

$$\hat{\sigma}_0^2 = \frac{1}{N} \sum_i (Y_i - \bar{Y})^2$$

Is this an unbiased estimator?

True vs. estimated sample distribution

For $Y_i \sim \text{Normal}(\mu, \sigma^2)$, the *true* sample distribution of $\hat{\mu} = \bar{Y}$ is

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But we don't know μ or σ ! In practice we plug in our estimates to get the estimated sample distribution

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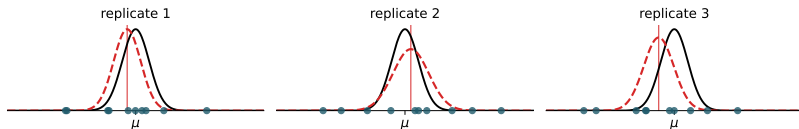
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Points: data ($N = 10$). Black: true. Red dashed: estimated, centered at $\hat{\mu}$.

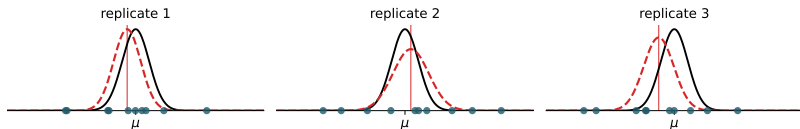
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We need to estimate the parameter in the sample distribution, which means the entire distribution is itself a random variable. Still, its width tells us how far μ might plausibly be from $\hat{\mu}$. This idea is formalized with confidence intervals.

Confidence intervals

The 95% confidence interval:

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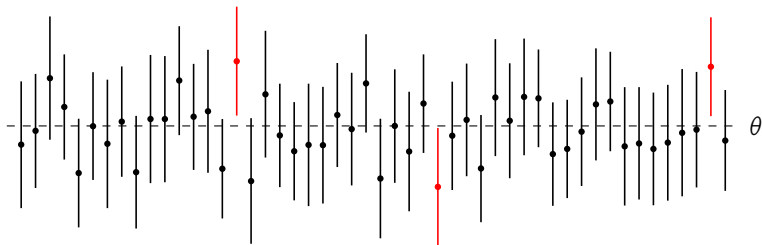
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Each bar is the CI from one replicate, with a dot at $\hat{\theta}$. About 95% overlap with θ .

Subtleties of interpretation (optional)

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- **Correct:** if we generate many replicates of the data, the true θ falls in the resulting CI for 95% of them; that is,

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- **Not correct:** there's a 95% chance θ lies in *this particular* interval; that is, say we compute $\text{CI} = [0.2, 1.4]$.

$$P(\theta \in \text{CI}) = 0 \quad \text{OR} \quad P(\theta \in \text{CI}) = 1$$

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$, where previous experiments give the estimate $\sigma \approx 1$, and estimate μ with $\hat{\mu} = \bar{Y}$. How large must N be so the 95%-CI, $\hat{\mu} \pm 1.96 \text{se}(\hat{\mu})$, is no wider than $\hat{\mu} \pm 0.25$?

The least squares estimator

Data $(X_1, Y_1), \dots, (X_N, Y_N)$ from $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$. How do we estimate β_1 ?

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($E[X]$ and $E[Y]$ are just numbers, so they come out of the expectation.)

Similarly $\sigma_X^2 = E[(X - E[X])^2]$.

The least squares estimator

Replace each expectation with a sample average, $E[\cdot] \rightarrow \frac{1}{N} \sum_i$, and $E[X], E[Y]$ with $\bar{X}, \bar{Y}$:

$$\text{cov}(X, Y) \approx \frac{1}{N} \sum_i (X_i - \bar{X})(Y_i - \bar{Y}), \quad \sigma_X^2 \approx \frac{1}{N} \sum_i (X_i - \bar{X})^2.$$

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This is the least squares estimator, since

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0, b_1}{\operatorname{argmin}} \sum_i r_i^2, \quad r_i = (Y_i - b_0 - b_1 X_i).$$

The terms r_i are called residuals.